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PAPER_988: REST F_U_Bi_i Endpoint — POST /api/fubi/master

Abstract

We document the 16th REST API endpoint added to `uqff_server.js` (Port 3141): `POST /api/fubi/master`. This endpoint accepts physical parameters (M, r, t, Γ) and returns the complete 6-layer F_{U,Bi_i} decomposition as JSON. It serves as the HTTP interface for all external consumers of the master buoyancy force.

1. Endpoint Specification

Property	Value
Method	POST
Path	`/api/fubi/master`
Port	3141
Content-Type	application/json

Request Body

```
{
    "M_kg": 1.989e30,
    "r_m": 1.496e11,
    "t_s": 86400,
    "Gamma_THz": 0.1
}
```

Response

```
{
    "F_U_Bi_i": -2.405685e-02,
    "layers": {
        "Ug_26layer": 5.930e-03,
        "Ub_26layer": -3.573e-03,
        "Um_magnetism": 1.234e-10,
        "UA_aether": -5.678e-12,
        "Fn_neutron": 1.0e-09,
        "Phi_phonon": 19.56,
        "E_net": -1.234,
        "S26": 19.56
```

```json
    },
    "metadata": {
        "SSq": 0.57,
        "beta_i": 0.603,
        "kappa_per_day": 0.0005,
        "omega_SCm_THz": 1.25,
        "layer_count": 6,
        "system_count": 1
    }
}
```

2. Error Handling

Status	Condition
200	Successful computation
400	Missing required fields or non-numeric values
500	Internal computation error (NaN/Inf guard)

3. Integration with uqff_server.js

The endpoint is the 16th route in the REST API server, complementing existing routes: - `/api/gravity/compressed` (POST) - `/api/gravity/resonance` (POST) - `/api/systems` (GET) - `/api/fubi/master` (POST) — **NEW** - … (12 others)

4. Usage Example

```
curl -X POST http://localhost:3141/api/fubi/master \
  -H "Content-Type: application/json" \
  -d '{"M_kg": 1.989e30, "r_m": 1.496e11, "t_s": 86400, "Gamma_THz": 0.1}'
```

5. Implementation

Route handler added to `uqff_server.js` — computes all 6 layers server-side using the same constants and equations as `fubi_master_calculator.py`.

References

- PAPER_979: Complete 6-Layer F_U_Bi_i
- PAPER_985: Production Kernel

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§A. Cosmogenesis-Linked Lagrangian

The REST endpoint makes the Lagrangian-derived F_{U,Bi_i} accessible to any HTTP client, enabling distributed computation across the 6-tier architecture.

§B. VDS/DVP/BSH Deep Synthesis

- **VDS:** The response `metadata.kappa_per_day` exposes the vacuum density growth rate to API consumers.
- **DVP:** Layer decomposition in the response reveals the DPM contribution via `Um_magnetism`.

- **BSH:** The `S26` field in the response is the buoyancy harmonic sum, central to all force calculations.